

Asymptotic enumeration and inversion for A122399

A proof and reproducible research report

1 October 2026

Abstract

For $a_n = \sum_k k^n k! S(n, k)$, we prove the leading equivalent already recorded in OEIS and obtain a complete fixed-order expansion with explicit coefficients. An exact vertical contour gives a branch-safe representation and a simple, nonasymptotic exponential truncation bound. The same method yields compact-direction two-size asymptotics, Gaussian block-count fluctuations, and a controlled inverse with integer-threshold envelopes. An elementary appendix proves the posted mod-prime periodicity conjecture, in the sense that $p - 1$ is a period, and extends it to prime powers. The standard saddle methodology and existing leading result are explicitly credited.

Results and attribution

The leading equivalent for A122399 was posted by Vaclav Kotesovec on OEIS on August 9, 2018; its growth constant was already identified in 2013. **Neither that equivalent nor its numerical constant is claimed as a new conjecture.** This manuscript proves it and develops explicit, reproducible higher coefficients, a globally valid contour with an explicit exponentially small truncation bound, a compact-direction extension, a block-count central limit theorem, and controlled inversion. These are applications and refinements of standard analytic saddle methods, not a new general method. In particular Khera–Lundberg–Melczer (2021, preprint 2019), Section 3 and the paragraph following its equation (15), give directional asymptotics and explain higher-order expansion for the nearby but different lonesum-matrix denominator. No global novelty claim is made.

Provenance, status and reading conventions

[write] **Added 2 October 2026, batch 77.** *Provenance.* This report was built from one manuscript: batch 77, manuscript 35 of the repository’s incoming-reports channel, archive `oeis-a122399-research.zip` (main file `a122399-report.tex`, 389 lines, with an eight-page PDF that is not shipped), arrival commit `096ee7b87`, placed in `f76fcb566`. The text from the abstract to the end of the source list is the delivered manuscript, unchanged except for the label prefix `a122:` and the notes marked [write]. The delivered pre-layout source `proof.md` is shipped beside this article; it carries no mathematics absent here, while Section 8 exists only in this text. *Pin.* The delivery names no ProveIt commit that it continues. Its `proof.md` (line 9) records a scoped search of the public repository index on 1 October 2026, which found no A122399 or A317855 match and returned other files at commit `76c0887a6`; the delivered layout script removed that paragraph from the manuscript. The search result was true then; the only match now is this report. The search is scoped nonduplication evidence, as its author says, not a proof of absence. *Status.* An unrefereed research manuscript; the delivery names no author or tool. Nothing in it is formalized in Lean or Rocq, and its placement in the collection confers no formal status (Section 7 names the one formalized identity its proof uses). The delivered independent audit (shipped as `audit-independent_audit.md`) found no blocking error; its script was not replayed at placement (it exceeded a 170-second cap), while the four producer scripts were replayed with identical outputs (up to line endings).

[write] **Reading conventions.** The manuscript reuses several letters. No symbol of the delivered text was renamed; the table fixes the readings.

symbol	meanings, by section
p	$p(z) = 1/(1 + e^z)$ in Sections 2–4; $p = x/y$ in the formula for μ_0 and $p(s)$ in Section 6; a prime in Section 7.
r	the Taylor index of f_r in (3.2), (3.4); the Newton step count in (5.3); the prime-power exponent in (7.1); the OEIS constant $r = 0.873702\dots$, which is $\mu = 1/y$ of (6.2), not any of the others.
x, y	formal variables of $F(x, y; u)$ in Section 1; saddle values $x = X(y)$, $y = y(\lambda)$ from (3.1) on, and on the diagonal the numbers $x(1)$, $y(1)$ of Section 4.
a	a_n is the sequence; a is a summation index in (3.4) and $a = y - x$ in (4.2).
m	the red set size (with $\lambda = m/n$); m_r in (3.4) are multiplicities.
C, C_J, E_J	C is the diagonal constant of (4.1); C_J and E_J in the proof of (5.4) are unrelated existence constants; $C(s)$ in Section 6 has $C(0) = C$.
μ, μ_0	mean of the block count (6.2), not a growth rate.
D	the growth constant of (4.1): $D = 1/(xy) = (1 + e^y)/y^2 = 3.16108865\dots$, the constant of OEIS A317855.

Section and equation numbers are the manuscript's own and are typed by hand; equation tags such as (2.1) are fixed tags, not counters.

1. Exact combinatorics

Put

$$A_{m,n} = \sum_{k=0}^m k^n k! S(m, k), \quad a_n = A_{n,n}, \quad b_{m,n} = \frac{A_{m,n}}{m!n!}.$$

The convention at the origin is $0^0 = 1$. For positive m, n , a structure consists of an ordered partition of an m -element red set into k nonempty blocks, followed by a map of an n -element blue set to these blocks. Its bivariate exponential generating function, with u marking blocks, is

$$F(x, y; u) = \frac{1}{1 - ue^y(e^x - 1)}.$$

Thus the diagonal is exactly A122399, not an approximation to it. When a structure is chosen uniformly, its block count satisfies $\Pr(K_n = k) = k^n k! S(n, k)/a_n$.

2. Exact contour and a nonasymptotic tail bound

For $\Re z > 0$, use the single analytic branch

$$X(z) = \text{Log}(1 + e^{-z}), \quad p(z) = \frac{1}{1 + e^z} = -X'(z).$$

Here $1 + e^{-z}$ stays in the open disk of radius one about 1, and $X(z) \neq 0$. The following identities hold for every pair of integers $m, n \geq 1$ and every $c > 0$:

$$\begin{aligned} b_{m,n} &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{p(z)}{X(z)^{m+1} z^{n+1}} dz \\ &= \frac{n+1}{m} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} X(z)^{-m} z^{-n-2} dz. \end{aligned} \tag{2.1}$$

Both integrals are absolutely convergent. Integer powers are used, so there is no unresolved power branch.

Proof

For $\Re h > 0$ and $n \geq 1$, the absolutely and locally uniformly convergent Mittag–Leffler identity is

$$\frac{1}{n!} \sum_{k \geq 1} k^n e^{-kh} = \sum_{j \in \mathbb{Z}} (h + 2\pi i j)^{-n-1}. \quad (2.2)$$

It follows, for example, by differentiating $1/(e^h - 1)$'s symmetric partial-fraction expansion; the constant term disappears after one derivative. It also follows by Poisson summation for $t^n e^{-ht} \mathbf{1}_{t \geq 0}/n!$, whose Fourier coefficients are the terms on the right.

Change the x -coefficient contour to $q = e^x - 1$, with $|q| = e^{-c} < 1$; the inverse is $x = \text{Log}(1 + q)$, analytic and one-to-one near the closed contour, and it winds once around zero. On this circle choose the continuous parametrization $q = e^{-c+it}$, $-\pi \leq t \leq \pi$, so $h = c - it$. Insert (2.2) in the Cauchy coefficient integral. Uniform absolute convergence permits the sum–integral interchange. The shifted intervals in $h + 2\pi i j$ join to the whole vertical line; X and p are $2\pi i$ -periodic. This gives the first identity in (2.1). Since $(X^{-m})' = mpX^{-m-1}$, integration by parts gives the second. Its boundary term tends to zero: on a fixed vertical line X is bounded away from zero, and $z^{-n-1} \rightarrow 0$. This also proves absolute convergence.

This argument does **not** treat an individual $(h(x) + 2\pi i j)^{-n-1}$ as a globally analytic power series at $x = 0$. Doing so would ignore its logarithmic monodromy.

Explicit truncation theorem

Let $x_c = X(c) > 0$, and let R_T denote the sum of the parts of the **second** integral in (2.1) with $|\Im z| \geq T > 0$, including its prefactor. Then

$$|R_T| \leq \frac{x_c^{-m} c^{-n}}{\pi m T} \left(\frac{c}{\sqrt{c^2 + T^2}} \right)^n. \quad (2.3)$$

Indeed, with $w = X(c + it)$,

$$e^{-c} = |e^w - 1| \leq e^{|w|} - 1, \quad \text{so} \quad |X(c + it)| \geq X(c).$$

Consequently the modulus of the tail is bounded by $(n + 1)x_c^{-m}/(\pi m) \int_T^\infty (c^2 + t^2)^{-(n+2)/2} dt$. Set $v = \sqrt{c^2 + t^2}$; since $v/\sqrt{v^2 - c^2}$ decreases, that integral is at most $(c^2 + T^2)^{-n/2}/((n + 1)T)$, proving (2.3).

There is therefore an exact, absolutely convergent strip decomposition obtained by splitting the second contour into $(2j - 1)\pi \leq \Im z \leq (2j + 1)\pi$, $j \in \mathbb{Z}$. The noncentral strips together have bound (2.3) with $T = \pi$. They are not asserted to be canonical saddle/transseries sectors: their artificial endpoint contributions can cancel, and no individual secondary-saddle expansion is proved here. A fixed-order Poincaré expansion for the central strip has algebraic error, despite the exponentially accurate contour truncation.

3. The unique saddle and all orders

Fix $\lambda = m/n > 0$. There is a unique $y = y(\lambda) > 0$ satisfying

$$x = X(y) = \lambda y p(y), \quad B = B(\lambda) = 1 + \lambda^{-1} - y(1 - p(y)). \quad (3.1)$$

Moreover $B > 1$, and x, y, B depend analytically on λ . To prove uniqueness, $R(y) = yp(y)/X(y)$ increases strictly from 0 to infinity. In fact $p/X > 1 - p$, since $p + (1 - p) \log(1 - p) > 0$ for $0 < p < 1$; differentiating $\log(p/X)$ shows its derivative is positive. Thus $R' > 0$. At the root $y(1 - p) < 1/\lambda$, proving $B > 1$. The implicit-function theorem supplies analyticity.

Define the analytic phase germ at $s = 0$

$$f_\lambda(s) = -\log(1 + s) - \lambda \log \frac{X(y(1 + s))}{x} = \frac{B}{2}s^2 + \sum_{r \geq 3} f_r s^r. \quad (3.2)$$

Here logarithms are local germs, and $f_\lambda(0) = f'_\lambda(0) = 0$. Globally along $s = i\theta$, the exponential in the following formula means its integer-power expression, so no global choice of f is needed:

$$b_{m,n} = \frac{x^{-m}y^{-n}}{\lambda y} \left(1 + \frac{1}{n}\right) \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{nf_\lambda(i\theta)}}{(1 + i\theta)^2} d\theta. \quad (3.3)$$

The bound from Section 2 gives $|e^{nf_\lambda(i\theta)}| \leq (1 + \theta^2)^{-n/2}$.

Explicit coefficient formula

Set $(-1)!! = 1$. For $j \geq 0$, sum over nonnegative integers $a, m_3, \dots, m_{2j+2}$ subject to

$$a + \sum_{r=3}^{2j+2} (r-2)m_r = 2j, \quad M = a + \sum_{r=3}^{2j+2} rm_r.$$

The integer M is necessarily even. Define

$$d_j = \sum (-1)^a (a+1) \prod_{r=3}^{2j+2} \frac{f_r^{m_r}}{m_r!} \frac{(-1)^{M/2} (M-1)!!}{B^{M/2}}, \quad c_0 = 1, \quad c_j = d_j + d_{j-1} \quad (j \geq 1). \quad (3.4)$$

The empty product gives $d_0 = 1$. For every fixed integer $J \geq 0$,

$$A_{m,n} = \frac{m!n! x^{-m}y^{-n}}{\lambda y \sqrt{2\pi n B}} \left(\sum_{j=0}^J \frac{c_j(\lambda)}{n^j} + O(n^{-J-1}) \right). \quad (3.5)$$

The remainder is uniform when m/n ranges over any fixed compact subset of $(0, \infty)$. No growing-direction or boundary-uniform assertion is made.

All-orders proof and remainder justification

Choose a small fixed $\delta > 0$ uniformly over the compact direction set. Outside $|\theta| < \delta$, the modulus bound above makes the integral exponentially small; integrate the power tail exactly or use (2.3). Inside, analyticity, $B > 1$, and compactness give uniform Taylor remainder bounds and a uniform quadratic bound $\Re f(i\theta) \leq -b\theta^2$, for some $b > 0$. Set $\theta = t/\sqrt{n}$. Expand the analytic amplitude $(1 + s)^{-2}$ and $\exp\{n(f(s) - Bs^2/2)\}$ through order $n^{-J-1/2}$, keeping enough further Taylor terms to bound the remainder. On $|t| \leq n^\epsilon$, with $\epsilon > 0$ chosen sufficiently small for this fixed J , Taylor's theorem bounds the remainder by n^{-J-1} times a fixed polynomial in $|t|$ times a Gaussian with uniformly negative exponent. On $n^\epsilon < |t| < \delta\sqrt{n}$, the uniform quadratic bound is superpolynomially small. Extend the polynomial–Gaussian integrals to the real line with the same accuracy. Odd terms

integrate to zero. The even Gaussian moments are $(M-1)!!B^{-M/2}$, with $i^M = (-1)^{M/2}$, giving (3.4). The prefactor $1 + 1/n$ gives $c_j = d_j + d_{j-1}$. This proves (3.5) with its uniform remainder.

This is the ordinary analytic one-dimensional saddle expansion; the exact contour is what supplies the particularly simple global tail control here.

4. Diagonal constants and reproducible coefficients

On the diagonal write $x = x(1)$, $y = y(1)$, $B = 2 - y + x$, $D = (xy)^{-1}$, and

$$C = \frac{1}{y\sqrt{2\pi B}}.$$

Numerically,

- $x = 0.2763928914715160816083833582415360977661$
- $y = 1.144554440911905470205106411159417937429$
- $B = 1.131838450559610611403276947082118160337$
- $D = 3.161088653865428813830172202588132491726$
- $C = 0.3276282855698694814422864924105070307103$

Thus

$$a_n = \frac{CD^n(n!)^2}{\sqrt{n}} \left[1 - \frac{0.0600744155748035800316371200380}{n} + \frac{0.00759452281787645507680871266458}{n^2} + \frac{0.00371322633571244961947936026057}{n^3} - \frac{0.000362980958541489832391365216986}{n^4} + O(n^{-5}) \right]. \quad (4.1)$$

An exact expression for the first coefficient is

$$c_1 = 1 - \frac{3}{B} + \frac{3f_4 - 6f_3}{B^2} - \frac{15f_3^2}{2B^3}, \quad (4.2)$$

where, setting $a = y - x$, $b = y - 2x$,

$$f_3 = \frac{a(b-3)}{6}, \quad f_4 = \frac{1}{2} - \frac{a}{2} + \frac{a^2}{8} + \frac{ab}{6} - \frac{a(y^2 - 6xy + 6x^2)}{24}.$$

The numerical generator uses exact rational symbolic algebra for the phase, then 90-digit arithmetic; it does not fit these coefficients to sequence values. It prints coefficients through n^{-7} , exact checks through $n = 800$, and scaled remainders. The recurrence $T_{m,k} = k(T_{m-1,k} + T_{m-1,k-1})$, $T_{0,0} = 1$, generates $T_{m,k} = k!S(m,k)$, so the numerical checks use independent exact integer values.

The strip-tail factor for $T = \pi$ is $\rho_\pi = y/\sqrt{y^2 + \pi^2} < 0.343$. Accordingly the exact central-contour integral differs from $a_n/(n!)^2$ by at most $D^n \rho_\pi^n / (\pi^2 n)$. This is a contour-evaluation statement, not an exponentially accurate bound for a fixed truncation of (4.1).

5. Controlled inverse

Define $P_J(t) = \sum_{j=0}^J c_j t^j$. For every fixed J , it is positive for all sufficiently small nonnegative t . For sufficiently large real v , put

$$H_J(v) = 2 \log \Gamma(v+1) + v \log D - \frac{1}{2} \log v + \log C + \log P_J(1/v).$$

Then

$$\log a_n = H_J(n) + O(n^{-J-1}), \quad H'_J(v) = 2 \log v + \log D + O(v^{-1}) > 0.$$

Consequently the large solution ν_J of $H_J(\nu_J) = \log a_n$ is unique and satisfies

$$\boxed{\nu_J = n + O(n^{-J-1}/\log n)}. \quad (5.1)$$

This follows from the mean-value theorem, using (3.5) and the derivative estimate. It is a controlled inverse along the sequence, without choosing an arbitrary exact interpolation of the integers.

For a practical Lambert-W initialization, put $L = \log A$ and

$$w = W_0\left(\frac{\sqrt{D}}{2e}L\right), \quad N_0 = \frac{L}{2w}.$$

If $A = a_n$, then $N_0 = n + O(1)$, and

$$n = N_0 - \frac{\frac{1}{2} \log N_0 + \log(2\pi C)}{2(w+1)} + O\left(\frac{1}{N_0 \log N_0}\right). \quad (5.2)$$

Indeed Stirling's expansion gives $\log a_n = 2n \log n + (\log D - 2)n + \frac{1}{2} \log n + \log(2\pi C) + O(1/n)$, while N_0 exactly solves its first two terms. Taylor expansion around N_0 gives (5.2).

[write] Added 2 October 2026, batch 77: an instance of repository results. The initializer of (5.2) is an instance of the inverse-Gamma core of the transseries-and-inversion volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, section `p6:sec:gamma`), and no novelty is claimed for the method. The two leading Stirling terms read $2N \log N + (\log D - 2)N = L$; with $T = \sqrt{D}N$ and $R = \sqrt{D}L/2$ this is the core equation $T(\log T - 1) = R$ of `p6:eq:gamma-core-eq`, whose solution $w = W_0(R/e)$, $T = R/w$ is $w = W_0(\sqrt{D}L/(2e))$, $N_0 = L/(2w)$ as above. The general power-logarithmic block is `p6:lem:core`, and the scaled family is `p2:thm:multi-scaling`. Only the core is shared: the lower-order terms here, $\frac{1}{2} \log N + \log(2\pi C)$, are not those of $\log \Gamma$, so the volume's all-orders recurrence for Γ_+^{-1} does not apply as it stands; this report's own all-orders inverse is ν_J of (5.1).

For arbitrary algebraic inverse precision use Newton iteration

$$N_{r+1} = N_r - \frac{H_J(N_r) - L}{H'_J(N_r)}, \quad (5.3)$$

starting at N_0 . Uniformly in its $O(1)$ starting neighborhood, $H''_J/H'_J = O((n \log n)^{-1})$. Therefore its error relative to ν_J after fixed r iterations is $O((n \log n)^{-(2^r-1)})$. Taking $2^r - 1 \geq J + 1$ and combining with (5.1) gives the inverse error in (5.1).

For an arbitrary large target A , let $\nu = H_J^{-1}(\log A)$ and $M(A) = \min\{n : a_n \geq A\}$. There is a constant $E_J > 0$ such that

$$\boxed{\left| \nu - E_J \frac{\nu^{-J-1}}{\log \nu} \right| \leq M(A) \leq \left| \nu + E_J \frac{\nu^{-J-1}}{\log \nu} \right|} \quad (5.4)$$

for all sufficiently large A . Indeed, the uniform sequence-value error $|\log a_k - H_J(k)| \leq C_J k^{-J-1}$ and the lower bound on H'_J show that integers below the lower real endpoint have $a_k < A$, and an integer at or above the upper endpoint has $a_k \geq A$. Enlarge E_J to cover both comparisons. The sequence is strictly increasing from $n = 1$ onward: its old positive block-count summands weakly increase and additional positive structures occur.

The envelope (5.4) allows two neighboring candidates near an integer boundary, so rounding a real inverse alone is not guaranteed. Exact integer comparison settles that boundary. The constants and starting threshold in (5.4) are asymptotic existence constants, not a certified finite-input error bar unless separately bounded.

[write] **Added 2 October 2026, batch 77.** The rounding remarks above are the general staircase facts of the same volume, `p0:thm:staircase`: part (1), $M(A) = \lceil \nu \rceil$ for the exact monotone interpolation, and part (2), the separation condition that fails near an integer boundary. Their generic arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`); those statements concern an arbitrary strictly monotone function and give no formal status to (5.1)–(5.4), whose interpolation H_J is not exact.

6. Block-count fluctuations

For $u = e^s$ near 1, replace X by $X_s(z) = \text{Log}(1 + e^{-s-z})$, and let $y(s)$, $x(s) = X_s(y(s))$ solve $x(s) = y(s)p(s)$, where $p(s) = 1/(1 + e^{s+y(s)})$. Set

$$B(s) = 2 - y(s)(1 - p(s)), \quad D(s) = [x(s)y(s)]^{-1}, \quad C(s) = \frac{1}{y(s)\sqrt{2\pi B(s)}}.$$

The all-orders theorem extends analytically and uniformly to a sufficiently small complex disk in s . To justify this rather than assuming it, use the analytic implicit-function theorem at the nondegenerate saddle. The contour can be locally deformed through $y(s)$ inside $\Re(s + z) > 0$. The fixed outer contour portions retain a strict exponential gap by continuity from $s = 0$, and the far tails retain uniform power bounds. The local analytic saddle expansion is therefore uniform in complex s . Cauchy estimates allow differentiation of its remainder.

It follows that

$$\mathbb{E}e^{sK_n} = \frac{C(s)}{C(0)} \left(\frac{D(s)}{D(0)} \right)^n [1 + O(n^{-1})], \quad (6.1)$$

uniformly near zero. This is the standard analytic quasi-powers situation. A direct expansion of the logarithm of (6.1) at $s = it/\sqrt{n}$, followed by the continuity theorem, gives the Gaussian limit stated below.

Let $\mu = 1/y$ and $\sigma^2 = (B - 1)/(By^2) > 0$. Then

$$\begin{aligned} \mathbb{E}K_n &= \mu n + \mu_0 + O(n^{-1}), & \frac{K_n - \mu n}{\sigma\sqrt{n}} &\Rightarrow N(0, 1), \\ \text{Var}(K_n) &= \sigma^2 n + O(1), \end{aligned} \quad (6.2)$$

where

$$\mu_0 = \frac{B - 1}{By} - \frac{B - 1 - p}{2B^2}, \quad p = x/y.$$

For verification, envelope differentiation gives $(\log D)' = 1/y$. Implicit differentiation of the saddle equation gives $y' = 1/B - 1$, hence $(\log D)'' = (B - 1)/(By^2)$. Further, $B' = (B - 1 - p)/B$ and $(\log C)' = -y'/y - B'/(2B)$, yielding μ_0 .

Numerically,

$$\begin{aligned}\mu &= 0.873702433239668330496568304721, \\ \sigma^2 &= 0.0889169868271170325484054553784, \\ \mu_0 &= 0.144565688577761026837729354227.\end{aligned}$$

This gives a probabilistic interpretation of the OEIS growth parametrization's constant $r = 0.873702\dots$: it is the asymptotic fraction of occupied ordered blocks. No local limit theorem or growing-marking-range result is claimed.

7. Elementary congruence appendix

The current OEIS entry conjectures that, for a prime p , the reduction of a_n for $n \geq 1$ is purely periodic with period $p - 1$. The following proves that $p - 1$ is a period; it does not assert that it is always the least period.

Proposition. For every prime p and integer $r \geq 1$,

$$a_{n+\varphi(p^r)} \equiv a_n \pmod{p^r} \quad (n \geq r). \quad (7.1)$$

In particular $a_{n+p-1} \equiv a_n \pmod{p}$ for every $n \geq 1$.

Proof. Terms with $k \geq pr$ vanish modulo p^r , since $v_p(k!) \geq \lfloor k/p \rfloor \geq r$. We may extend the sum up to $pr - 1$ even when this exceeds n , since $S(n, k) = 0$ for $k > n$. The exact inclusion-exclusion formula gives

$$a_n \equiv \sum_{k=0}^{pr-1} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} (kj)^n \pmod{p^r}.$$

For $n \geq r$, a term with $p \mid kj$ is zero modulo p^r . Each other base kj is a unit, so Euler's theorem leaves its power unchanged on replacing n by $n + \varphi(p^r)$. This proves (7.1). The argument is elementary and independent of the asymptotic analysis.

[write] **Added 2 October 2026, batch 77.** The OEIS conjecture is credited there to Peter Bala (31 May 2022), as the delivered `source_review.md` records. Neither (7.1) nor any other statement of this report is formalized. The one nontrivial identity in its proof, the surjection formula $k!S(n, k) = \sum_{j \leq k} (-1)^{k-j} \binom{k}{j} j^n$, is formalized in Lean as `Fabius.factorial_mul_stirlingSecond_eq_sum` (`Analysis/FabiusFunction/Lean/FabiusFunction/StirlingBasisChange.lean`), so (7.1) is a small formalization target on top of it and Euler's theorem; no such proof exists in the repository. The collection's report `congruences-and-valuations/secant-number-periodicity` studies the same kind of question for the secant numbers A000364, where the analogous OEIS conjecture (period dividing $\varphi(m)$, pure from index one) is false for some moduli; no result is shared, and no least period or onset is claimed here.

8. Questions left by this analysis

The results separate three different levels of control: an exact contour with an explicit exponential tail, a fixed-order algebraic saddle expansion, and an asymptotic inverse. Several natural extensions require additional arguments.

1. **Exponentially improved expansions.** Locate and classify the complex saddles of $z \operatorname{Log}(1 + e^{-z})$, determine which are reached by valid contour deformations, and obtain an optimally truncated expansion with a controlled secondary-saddle remainder. The exact strip decomposition alone does not answer this question.

2. **Directions approaching the boundary.** Determine uniform laws when $m/n \rightarrow 0$ or $m/n \rightarrow \infty$, including the transition scales where the compact-direction expansion ceases to be uniform. These regimes may require different saddle coordinates and block-count normalizations.
3. **Effective finite-input inverse bounds.** Turn the implicit all-orders constants into computable numerical bounds and starting thresholds, combining analytic Taylor estimates with the explicit contour tail. This would make the ceiling envelope a certified finite-input bracket and could reduce the need for exact comparisons away from sequence-value boundaries.
4. **Sharper block-count laws.** Prove a local limit theorem, explicit Edgeworth terms, or a uniform large-deviation estimate for the block count. The analytic marking germ provides the central cumulants, but the required global Fourier or tilted-contour estimates are not supplied here.

These are limitations and possible continuations of this analysis, not claims that the corresponding general techniques or all related results are absent from the literature.

Sources and reproducibility

1. OEIS A122399, current entry inspected 2026-10-01: <https://oeis.org/A122399>
2. OEIS A317855: <https://oeis.org/A317855>
3. Jessica Khera, Erik Lundberg, Stephen Melczer, *Asymptotic Enumeration of Lonesum Matrices*, Advances in Applied Mathematics 123 (2021), article 102118, doi:10.1016/j.aam.2020.102118; arXiv:1912.08850: <https://arxiv.org/html/1912.08850> (especially Section 3, Theorem 3.1, and higher-order comment after equation (15)). Its denominator is $e^{-x} + e^{-y} - 1$, different from the one here.
4. `check_a122399.py`: exact rational phase generation, finite-sum saddle coefficients, exact-integer comparisons, and moment checks; outputs `numerical_results.json`, `phase_coefficients.txt`, and `test_output.txt`.
5. `export_exact_coefficients.py`: exact rational expressions through the fourth correction, including fully combined rational functions of x, y for the first two corrections; the exporter asserts that no floating-point atoms occur.
6. `verify_contour_inverse.py`: independent direct quadrature of the finite vertical contour with the explicit tail certificate, and inverse-iteration checks.
7. `verify_congruences.py`: finite modular checks of (7.1), including all primes below 50 and several prime powers.

[write] **Added 2 October 2026, batch 77.** *Item 2.* The manuscript lists OEIS A317855 without saying what it is. Checked on 2 October 2026: A317855 is the decimal expansion 3.161088653... of the growth constant D of (4.1) (OEIS titles it as a constant of the asymptotics of A122400), and the A122399 entry writes the Kotěšovec equivalent as $c \cdot A317855^n (n!)^2 / \sqrt{n}$ with $c = 0.32762828... = C$, its constant r being $1/y$. The manuscript states no other relation between A317855 and A122399. *Items 4–7 and the audit.* The scripts are shipped as `code/check_a122399.py`, `code/export_exact_coefficients.py`, `code/verify_contour_inverse.py` and `code/verify_congruences.py`, their recorded outputs under `data/`, and the audit as `audit-independent_audit.md`, `code/audit-independent_audit.py` and `data/audit-independent_audit.*`. They read and write files next to themselves in the delivered flat layout; the report’s README gives rerun instructions that do not overwrite shipped data. *Neighbours.* Row $n = 0$ of $A_{m,n}$ is the Fubini sequence, $A_{m,0} = \sum_k k! S(m,k)$, defined in Lean as `Fabius.fubini` (`Analysis/FabiusFunction/Lean/FabiusFunction/OrderedBell.lean`); its exact pole-lattice formula is `q2.thm:fubini` of the transseries volume. That row lies outside the compact-direction theorem (3.5), on the side of question 2 of Section 8, and nothing is claimed about it here. The collection’s `a277364-bell-asymptotics` treats a

different Stirling sum by classical saddle methods; no theorem is shared.

The computer checks are reproducible evidence; the all-orders and error statements rely on the analytic proofs above. An independent audit and clean replay accompany this report. Conventional peer review remains appropriate before publication.